

Candidate Name

Centre Number

Candidate Number



For Performance Measurement

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Ordinary Level

COMBINED SCIENCE

PAPER 2 Theory

4003/2

2 hours

SPECIMEN PAPER

Additional materials:
 Calculator (Optional)
 Answer booklet

INSTRUCTIONS TO CANDIDATES

Write your name, centre number and candidate number in the spaces at the top of this page.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question.

This question paper consists of **four** compulsory sections, **A, B, C and D**.

The Periodic Table is provided on Page 15.

FOR EXAMINER'S USE

Section A**SECTION B****Section C****Section D****TOTAL**

This question paper consists of 15 printed pages.

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Section A

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- 1 (a) Fig 1.1 shows the human respiratory system.

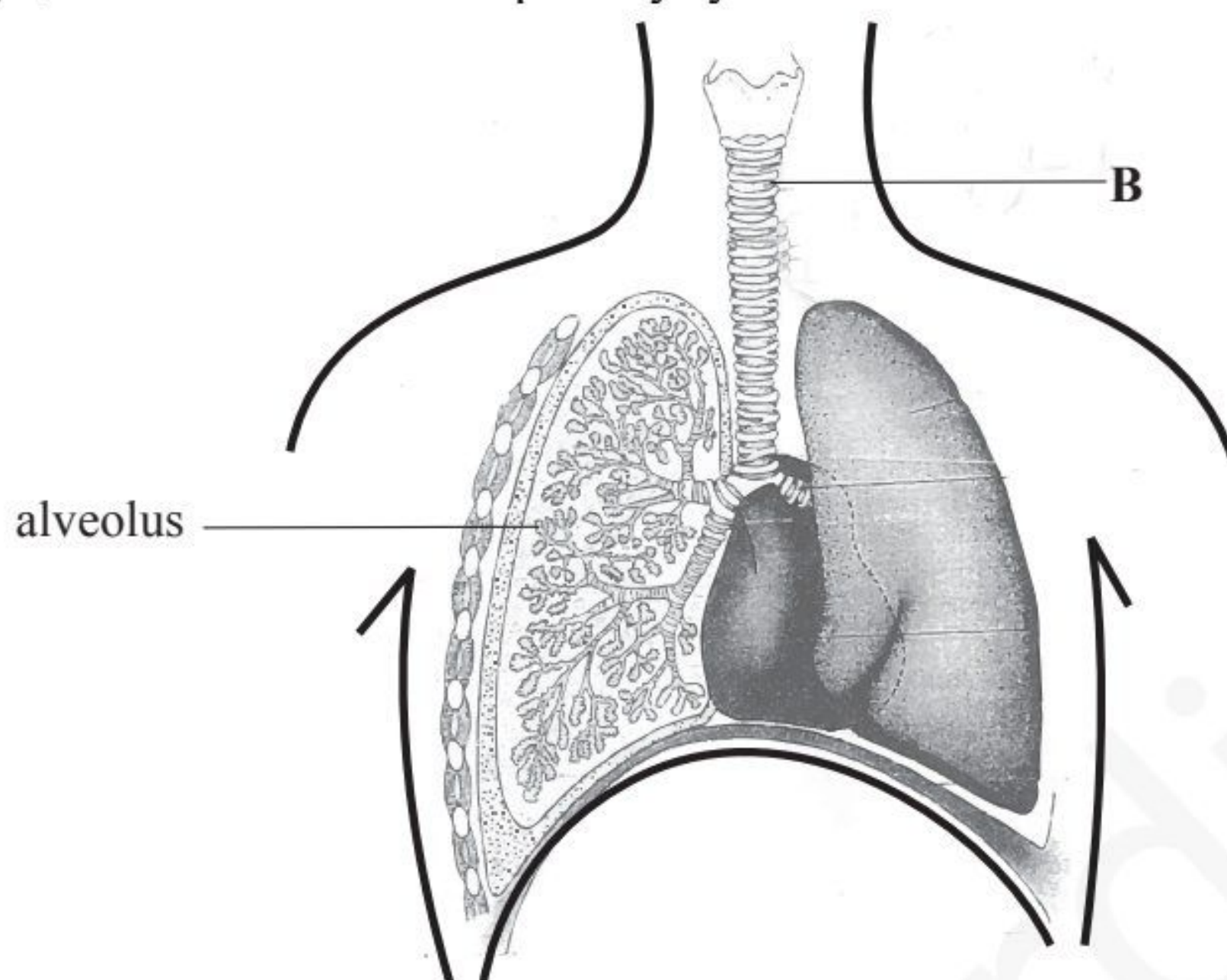


Fig. 1.1

- (i) Name the part labelled **B**.

_____ [1]

- (ii) State any **one** feature of the named part that enables it to clean the air.

_____ [1]

- (b) State the function of the alveolus.

_____ [1]

- (c) (i) Name the inspired gas used in respiration.

_____ [1]

- (ii) Describe a positive chemical test for the gas stated in (c)(i).

[2]

- (d) State the gas whose percentage concentration remains the same during breathing in and breathing out.

[1]

- 2 (a) Fig. 2.1 shows an agent of sugar bean pollination.

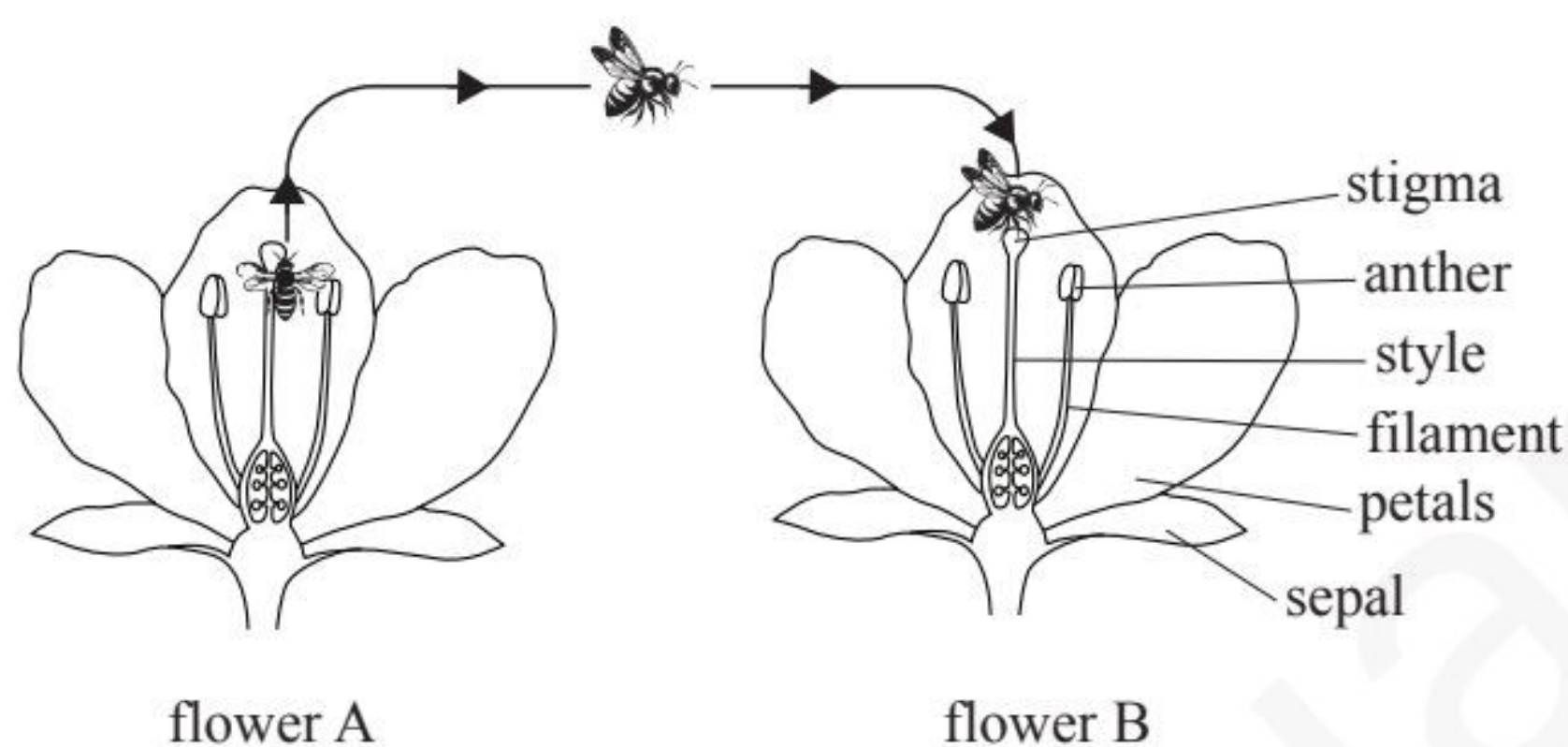


Fig. 2.1

- (i) Define the term *pollination*.

[1]

- (ii) State the type of pollination shown in Fig. 2.1.

[1]

- (b) Contact insecticides are chemicals which kill insects.
The chemicals can remain active for more than 14 days after application.

- (i) Explain the effect of using a contact insecticide on a crop of sugar beans at flowering stage.

[2]

- (ii) Describe how fertilisation takes place in a sugar bean flower.

[2]

- (iii) Describe the change which occurs to the ovary after fertilisation.

[1]

- 3 (a) A learner washed and placed an iron hoe in a damp shade. After three days, a brown substance had formed on the hoe.

- (i) Identify the process responsible for the formation of the brown substance.

[1]

- (ii) State **two** conditions necessary for the process identified in (a)(i) to occur.

[2]

- (iii) Suggest an improvement that can be done to the shade in order to reduce the occurrence of the process identified in (a)(i).

[1]

- (b) A manufacturer decides to supply hoes that are made up of galvanised iron.

- (i) Explain how the galvanized hoe would resist the process identified in (a)(i).

[2]

- (ii) Suggest **one** disadvantage of galvanising the hoe.

_____ [1]

- 4 A dilute acid and a metal react to produce a salt and a gas.

- (a) (i) Name the gas produced.

_____ [1]

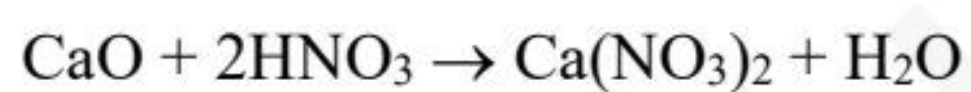
- (ii) State the type of bonding in the gas.

_____ [1]

- (iii) Identify any **one** industrial use of the gas.

_____ [1]

- (b) Calcium oxide and nitric acid react to form a salt and water as shown in the equation.



- (i) Name the salt produced.

_____ [1]

- (ii) Calculate the mass of one mole of the salt produced.

[3]

- 5 (a) (i) Define the term *energy*.

_____ [1]

- (ii) Describe how sound is produced.

_____ [1]

- (b) Fig. 5.1 shows two cellphones in two jars.

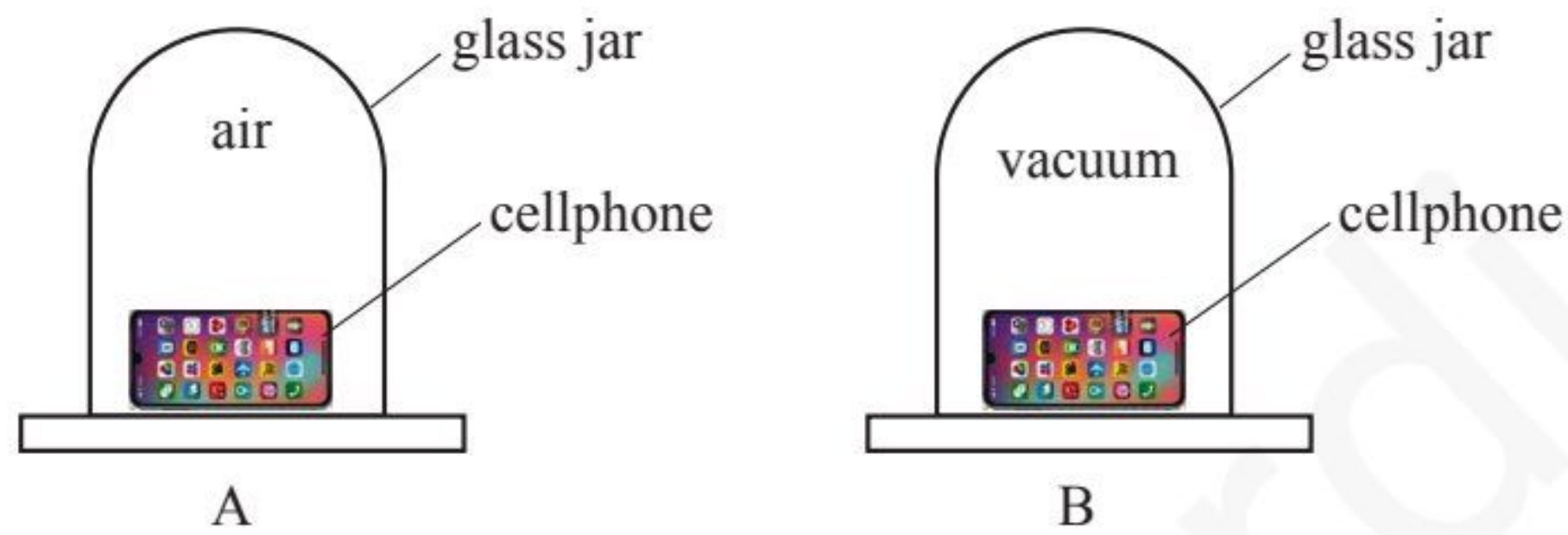


Fig. 5.1

- (i) Describe how the sound of the cellphone in **A** reaches the owner when it rings.

_____ [2]

- (ii) Explain why the sound of the cellphone in **B cannot** be heard when it rings.

_____ [1]

- (c) Fig. 5.2 shows an incomplete light ray diagram.

Complete the diagram in Fig. 5.2 to show the formation of a shadow.

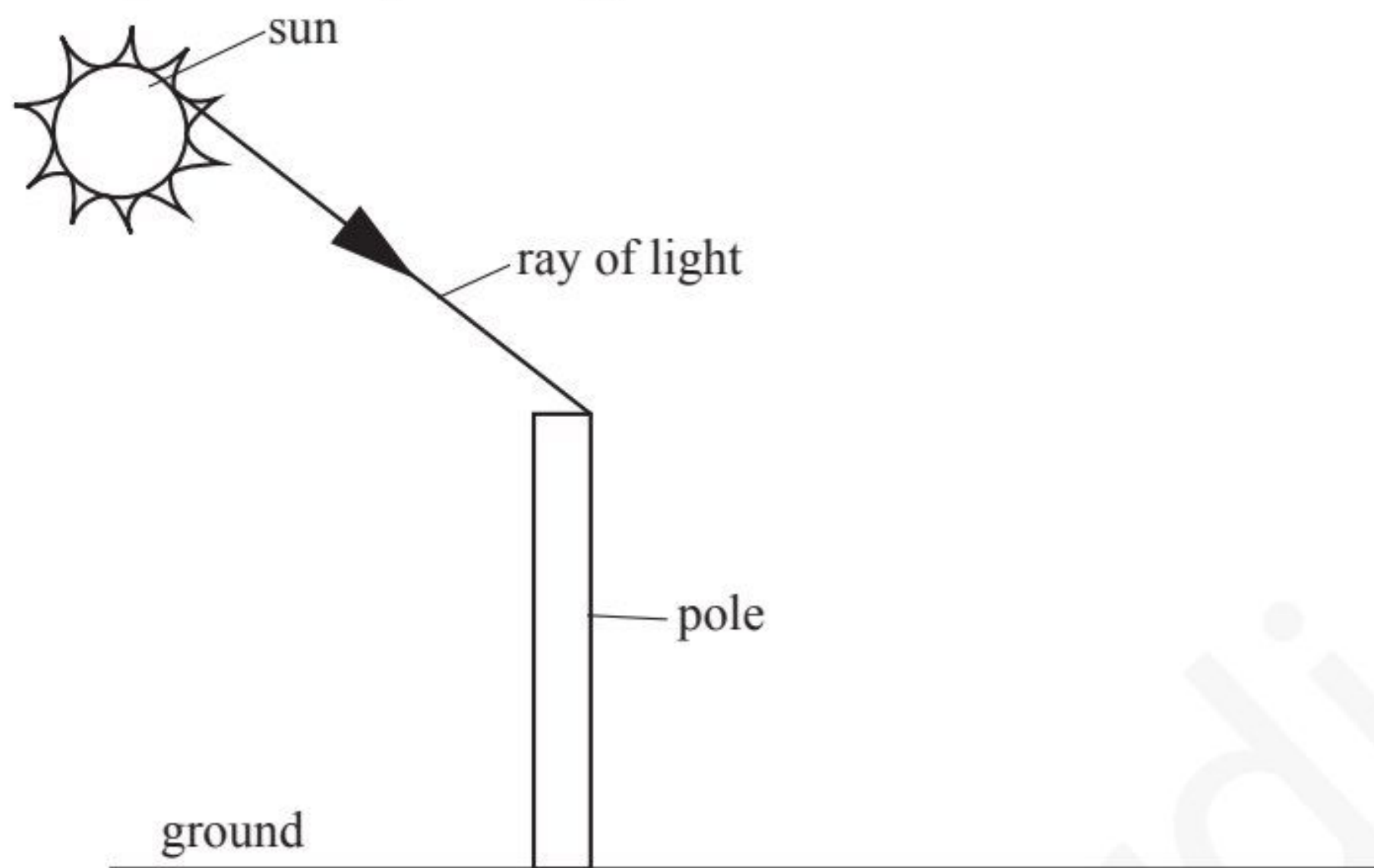


Fig. 5.2

- 6 (a) (i) Define the term *telecommunication*.
 _____ [1]
- (iii) Long ago, kudu horns and drums were used for communication in Zimbabwe.
 Identify any **two** modern ways used for communication in Zimbabwe.
 1. _____
 2. _____ [2]
- (b) Explain why a cellphone can be described as a transceiver.

 _____ [1]

- (c) Fig. 6.1 shows a communication medium.

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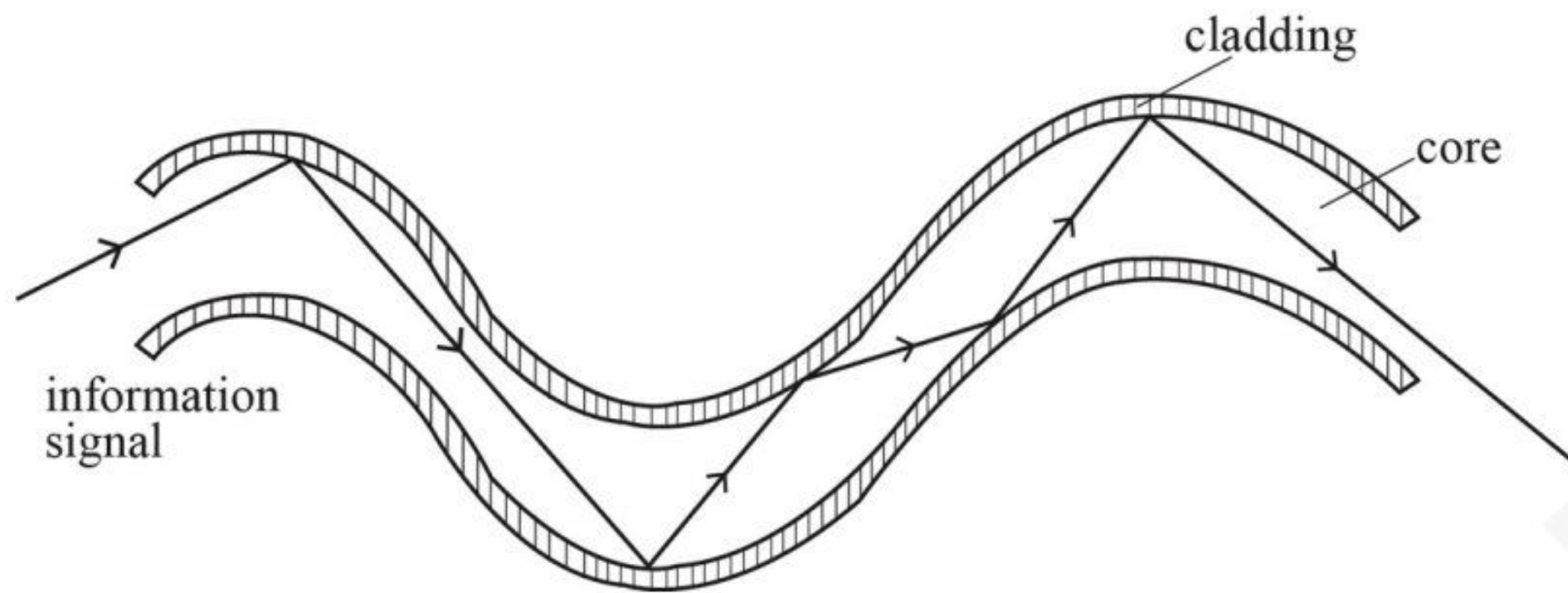


Fig. 6.1

- (i) Identify the medium.

[1]

- (ii) Outline how messages are transmitted through the medium named in (c)(i).

[2]

Section B

Answer any **two** questions in the answer booklet provided.

- 7 (a) (i) Write the word equation for photosynthesis. [2]
- (ii) State any **one** use of each of the products of photosynthesis. [2]
- (b) (i) Describe any **two** leaf adaptations for photosynthesis. [2]
- (ii) Explain the adaptations described in (c)(i). [2]
- (c) Name any **two** food storage organs in a plant. [2]

- 8 Fig. 8.1 shows how the rhythm method is related to the menstrual cycle.

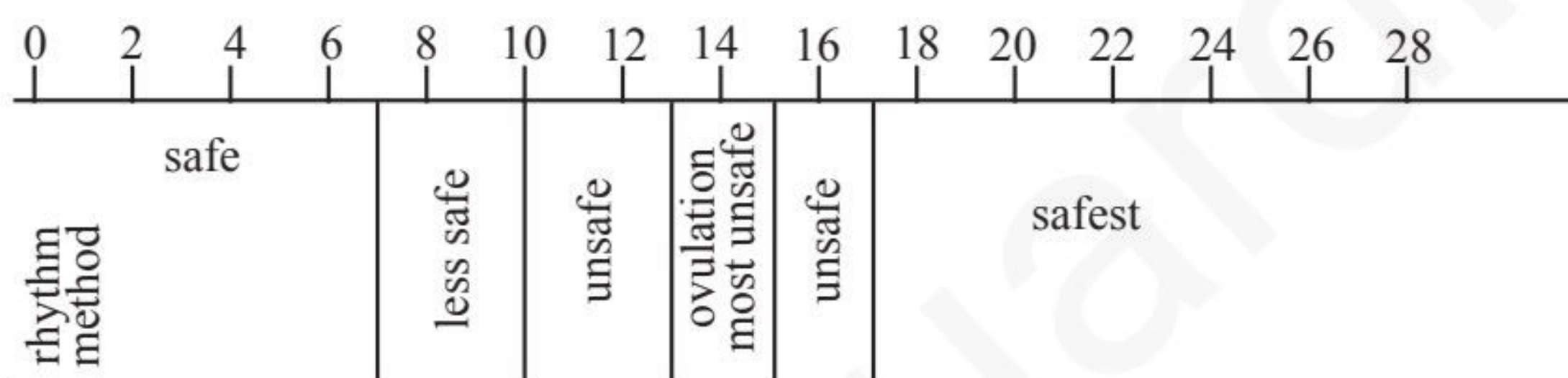


Fig. 8.1

- (a) (i) Name any **one** hormone which controls the menstrual cycle. [1]
- (ii) Explain why the ovulation period is the most unsafe period. [2]
- (iii) Explain why the period between days 10 and 13 is unsafe. [2]
- (b) (i) State any **two** advantages of using the rhythm method. [2]
- (ii) Give any **one** other natural method of contraception. [1]
- (c) The use of a condom falls under barrier methods.
- (i) Describe how the use of the condom prevents fertilisation. [1]
- (ii) State **one** other advantage of using the condom apart from preventing fertilisation. [1]

- 9 (a) Most low lying areas in Zimbabwe are still prone to malaria. The malaria vectors thrive in these areas.

Identify the malaria vector.

[1]

- (b) Fig. 9.1 shows the stages in the life cycle of the malaria vector.

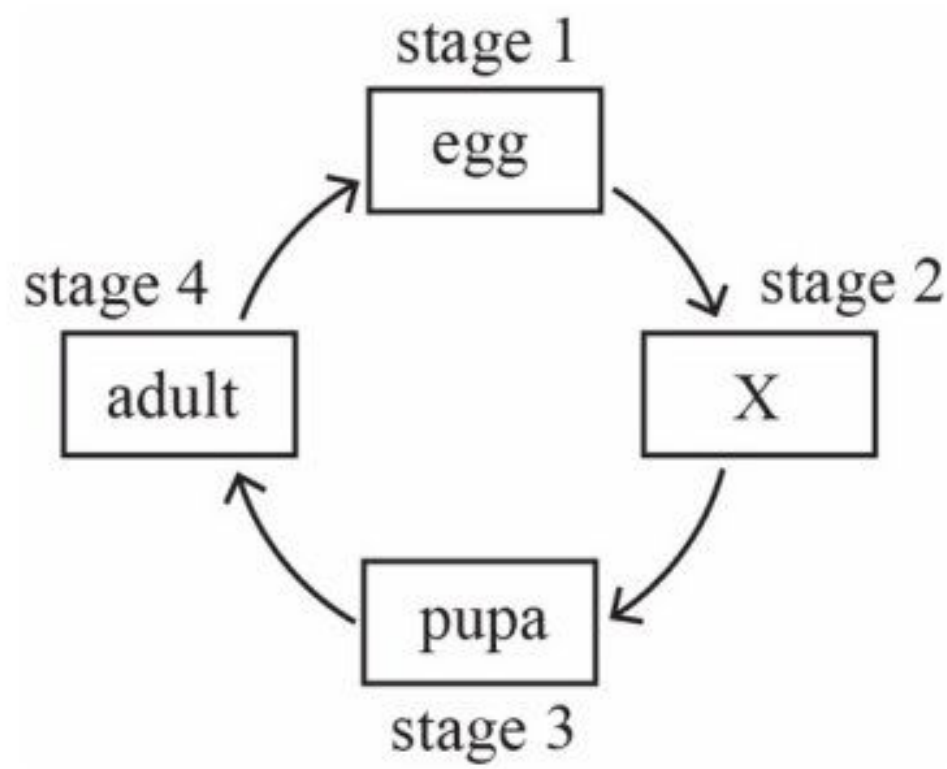


Fig. 9.1

- (i) Name the stage of the cycle represented by X. [1]
- (ii) Suggest any **two** methods that can be used to break the cycle at
1. stage 1,
 2. stage 4.
- [4]
- (iii) State any **two** signs or symptoms of malaria. [2]
- (c) Explain effects of malaria to a community. [2]

Section C

Answer any **two** questions in the answer booklet provided.

- 10 **Fig. 10.1** shows the apparatus used to separate two miscible liquids, A (boiling point 120°C) and B (boiling point 60°C).

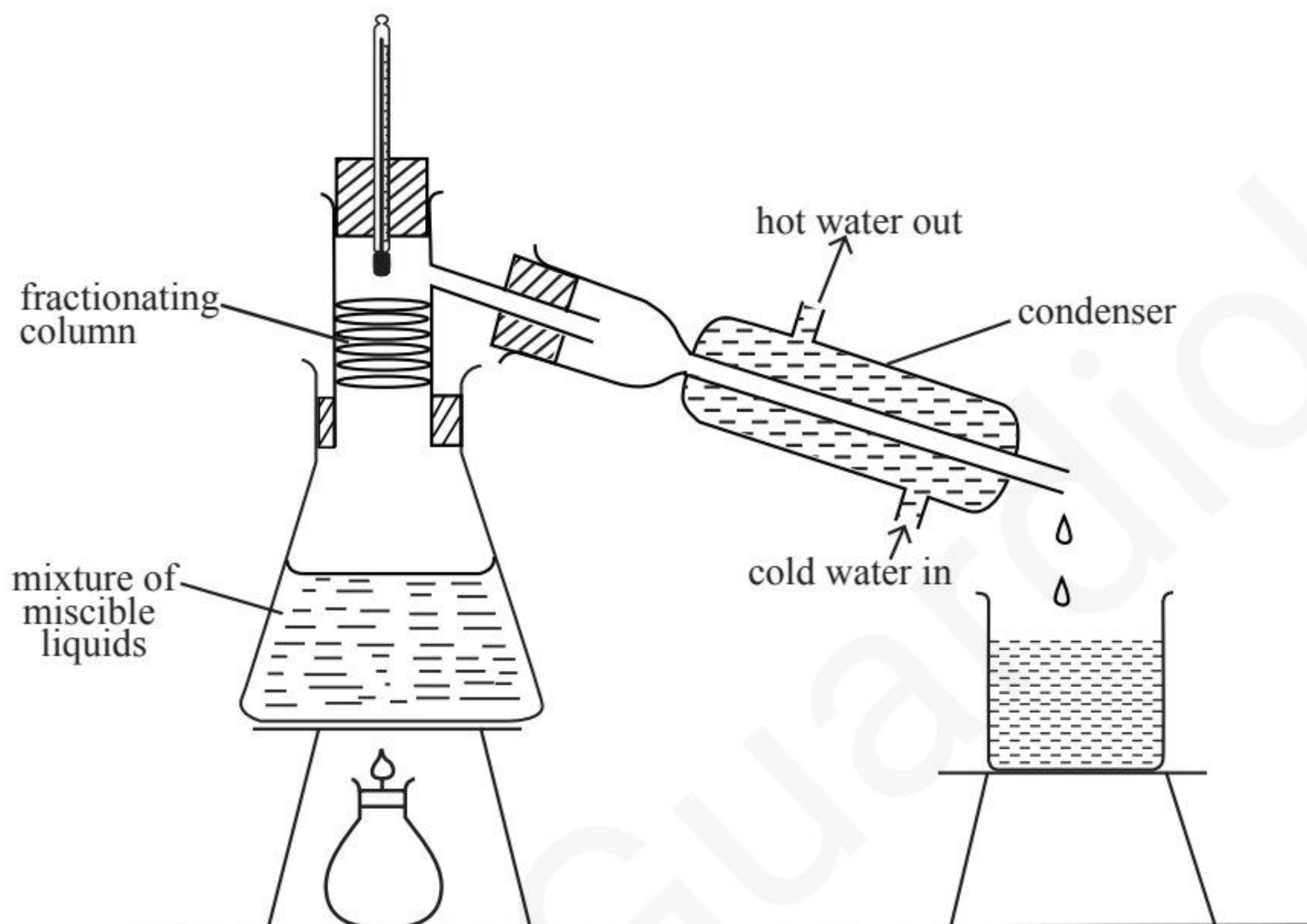


Fig. 10.1

- (a) Identify the method of separation shown in **Fig. 10.1**. [1]
- (b) (i) Describe and explain how the two liquids are separated. [4]
- (ii) Explain why cold water gets into the condenser through the lower opening. [2]
- (c) Explain any **one** industrial application of the method shown in **Fig. 10.1**. [3]

- 11 Table 11.1 shows electronic configurations of elements represented by letters A to F.

Table 11.1

Element	Electronic configuration
A	2.1
B	2.8.2
C	2.8.7
D	2.6
E	2.8.1
F	2.8

- (a) State and explain using letters, A to F,
- (i) **two** elements which are in the same Group of the Periodic Table, [2]
 - (ii) any **one** element which forms a negative ion, [2]
 - (iii) an element which is unreactive. [2]
- (b) (i) Describe how chemical bonding occurs between elements C and E. [3]
- (ii) State the proton number of the element B. [1]
- 12 Fig. 12.1 shows displayed formula of an organic compound.

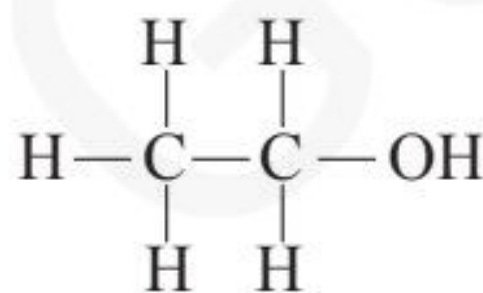


Fig. 12.1

- (a) (i) Name the organic compound. [1]
- (ii) State the homologous series to which the compound belongs. [1]
- (iii) Give any **one** use of the compound. [1]
- (b) A hydrocarbon contains 86% carbon and 14% hydrogen by mass.
- (i) Calculate the empirical formula of the hydrocarbon. [3]
 - (ii) Determine the molecular formula of the hydrocarbon, given that the relative molecular mass is 42. [2]
 - (iii) State the **two** products of the complete combustion of the hydrocarbon. [2]

Section D

Answer any **two** questions in the answer booklet provided.

- 13 (a) (i) Name the instrument used to measure force. [1]
- (ii) State the Standard International (SI) unit of force. [1]
- (b) (i) Define the term *moment of a force*. [1]
- (ii) **Fig. 13.1** shows a spanner being used to undo a rusty nut from the wheel of a scotch cart.

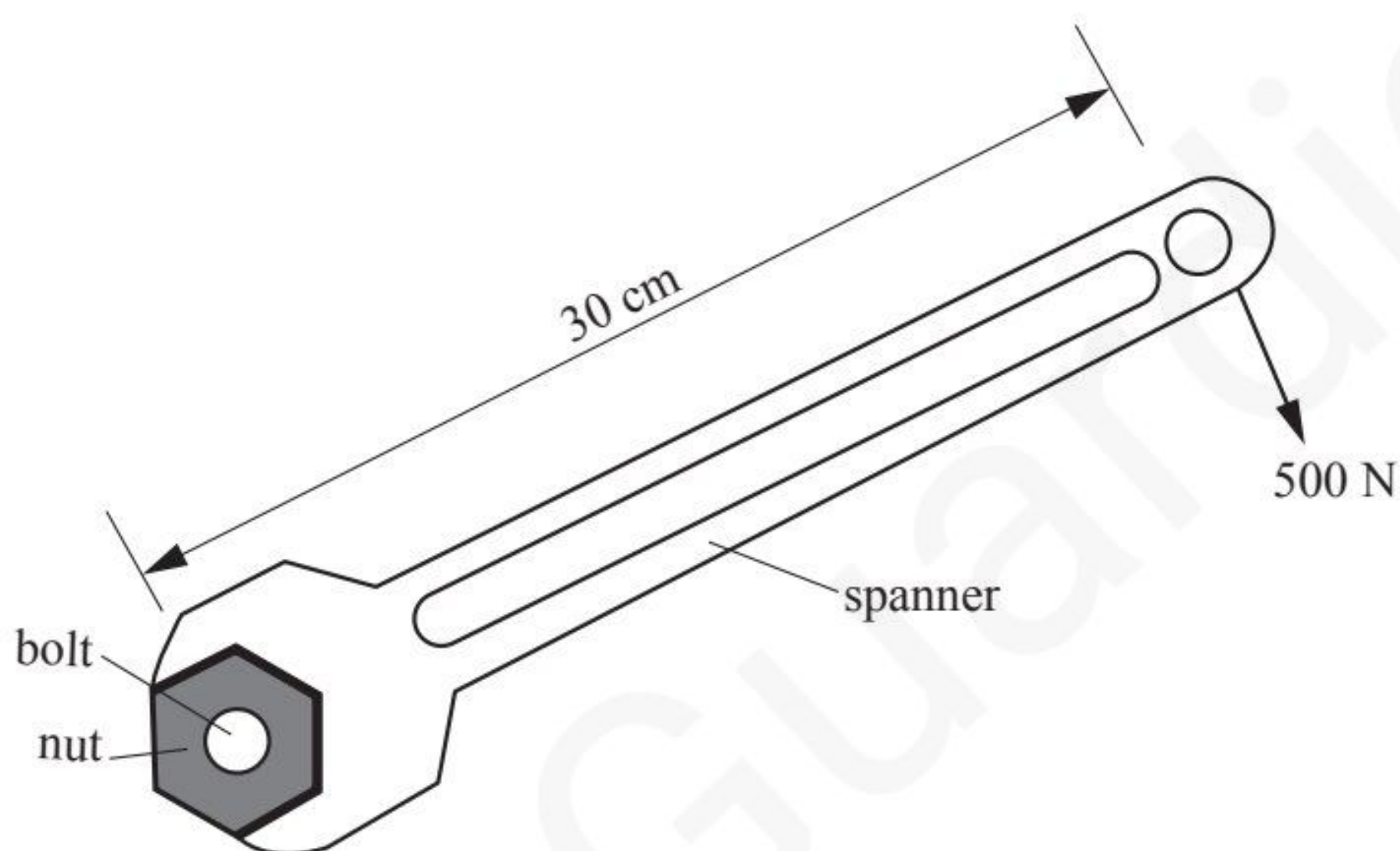


Fig. 13.1

- Calculate the moment generated. [3]
- (iii) Explain the effect of the rust between the nut and the bolt on the amount of effort applied. [2]
- (d) Describe **two** ways of making the removal of the nut easier. [2]
- 14 (a) List any **two** uses of electricity in the home. [2]
- (b) (i) State any **two** safety features in a three pin plug. [2]
- (ii) Explain how any **one** of the safety features stated in (b)(i) works. [1]

- (c) **Table 14.1** shows Zimbabwe Electricity Supply Authority (ZESA) electricity tariffs for a certain period of time.

Table 14.1

electricity tariffs	
energy/kWh	cost per unit
1 – 50	4 cents
51 – 100	7 cents
101 – 200	9 cents
201 – 300	12 cents
301 – 400	14 cents

- (i) Calculate the cost of buying 250 kWh of energy. [3]
- (ii) Give any **two** ways of saving electricity in the home. [2]
- 15** (a) Give any **three** components of a robot. [3]
- (b) Describe any **two** uses of fixed robots. [2]
- (c) Explain why a **NOT** gate is also called an inverter. [2]
- (d) Outline the stages in robot production. [3]

DATA SHEET

The Periodic Table of the Elements

Group																	
I	II	1 H Hydrogen 1										III	IV	V	VI	VII	O
7 Li Lithium 3	9 Be Beryllium 4											11 B Boron 5	12 C Carbon 6	14 N Nitrogen 7	16 O Oxygen 8	19 F Fluorine 9	20 Ne Neon 10
23 Na Sodium 11	24 Mg Magnesium 12											27 Al Aluminium 13	28 Si Silicon 14	31 P Phosphorus 15	32 S Sulphur 16	35.5 Cl Chlorine 17	40 Ar Argon 18
39 K Potassium 19	40 Ca Calcium 20	45 Sc Scandium 21	48 Ti Titanium 22	51 V Vanadium 23	52 Cr Chromium 24	55 Mn Manganese 25	56 Fe Iron 26	59 Co Cobalt 27	59 Ni Nickel 28	64 Cu Copper 29	65 Zn Zinc 30	70 Ga Gallium 31	73 Ge Germanium 32	75 As Arsenic 33	79 Se Selenium 34	80 Br Bromine 35	84 Kr Krypton 36
85 Rb Rubidium 37	88 Sr Strontium 38	89 Y Yttrium 39	91 Zr Zirconium 40	93 Nb Niobium 41	96 Mo Molybdenum 42	101 Ru Ruthenium 44	103 Rh Rhodium 45	106 Pd Palladium 46	108 Ag Silver 47	112 Cd Cadmium 48	115 In Indium 49	119 Sn Tin 50	122 Sb Antimony 51	128 Te Tellurium 52	127 I Iodine 53	131 Xe Xenon 54	
133 Cs Caesium 55	137 Ba Barium 56	139 La Lanthanum 57	178 Hf Hafnium 72	181 Ta Tantalum 73	184 W Tungsten 74	186 Re Rhenium 75	190 Os Osmium 76	192 Ir Iridium 77	195 Pt Platinum 78	197 Au Gold 79	201 Hg Mercury 80	204 Tl Thallium 81	207 Pb Lead 82	209 Bi Bismuth 83	210 Po Polonium 84	210 At Astatine 85	222 Rn Radon 86
226 Fr Francium 87	226 Ra Radium 88	227 Ac Actinium †															

*58-71 Lanthanoid series
†90-103 Actinoid series

a = relative atomic mass

b = proton (atomic) Number

a

X

X = atomic symbol

The volume of one mole of any gas is 28 dm^3 at room temperature and pressure (r.t.p.)